

Babalola John Abidemi

+2349168881444 | Ogbomoso, Nigeria | abidemibabalola23@gmail.com | [Linkedin](#) | [Github](#) | [Portfolio](#)

EDUCATION

Ladoke Akintola University of Technology (LAUTECH), Ogbomoso, Oyo State
B.Tech in Electronic and Electrical Engineering CGPA: **4.70/5.0 (3.8/4.0) WES**

Dec. 2021 - Aug 2026

Software Skills: Python, C++, Robot Operating System (ROS2- SLAM, NAV2, Path Planning, Teleoperation, ROS - AI Integration), Agentic AI, MATLAB/Simulink, Pytorch, Reinforcement Learning and Imitation Learning.

Hardware Skills: Embedded System, Arduino Programming, CAD/CAM Design (Fusion360), hardware programming (Raspberry pi, Arduino, ESP32), Laser Cutting/Engraving.

Research Interest: Robot Perception and Autonomous System, Human Robotic Interaction (HRI), Space and Field Robotics, Embodied AI and Agent-based Robot Architecture, Sensor Fusion and Edge AI for Autonomous Systems, Autonomous Navigation.

RESEARCH AND WORK EXPERIENCE

Undergraduate Researcher

Unilag Design Studio University of Lagos, Lagos

May. 2025 - Oct. 2025

- Designed an AI-powered pick-and-place robot that translates natural language commands into autonomous manipulation in confined spaces using ROS 2 (MoveIt 2), computer vision and agentic AI.
- Developed **VisionLock**, a high-performance embedded AI security system leveraging computer vision-based identity recognition for autonomous access control.
- Designed and implemented **CourierX**, a digital-twin enabled, AI-powered autonomous delivery robot integrating ROS, LiDAR-based perception, SLAM and autonomous navigation for safe and reliable last-mile delivery.
- Developed a comprehensive, beginner-friendly tutorial covering Convolutional Neural Networks (CNNs), PyTorch and computer vision fundamentals.
- Architected an end-to-end AI system integrating research automation, web scraping pipelines, intelligent task orchestration and AI-powered home automation services within a unified framework

Student Industrial Work Experience Scheme

Unilag Design Studio University of Lagos, Lagos

Mar. 2025 - May. 2025

- Received hands-on training in embedded systems, Arduino programming and IoT architectures applied to robotics and intelligent systems.
- Trained in project management methodologies and the art of producing clear, structured technical documentation.
- Led a team of five to design and develop a ROS-based fully autonomous mobile robot (AMR) for intelligent navigation and learning applications.

Embedded Systems Intern

Elord Technology, Ogbomosho, Lagos

May. 2024 - May. 2025

- Implemented infrared sensor configurations for closed-loop solar tracking systems, enabling automated panel alignment and improved energy capture.
- Developed **Attendify**, an RFID-enabled smart attendance system integrating embedded hardware with cloud-based data logging and automation via Google Sheets.

Student Work Experience Program

Ladoke Akintola University of Technology, Electronic/Electrical Department,

Feb. 2024 - May. 2024

- Received foundational training in basic electronics and electrical circuitry.
- Trained in writing comprehensive and well-structured technical project reports.

EXTRA-CURRICULAR/LEADERSHIP EXPERIENCE

Institute of Electrical and Electronic Engineering LAUTECH Student Branch | Ogbomosho.

Project Coordinator |

Mar. 2025 - Present

- Led and facilitated an impactful 3-day embedded systems bootcamp for undergraduates, introducing Arduino programming and the application of embedded systems in robotics.
- Hosted a technical session on bridging the gap between semiconductor technologies and real-world industrial applications.

- Led and coordinated student projects and participated in writing technical grant applications for robotics and environmental sustainability initiatives.

Low Budget Engineer | Ibadan, Oyo State.

Exhibition and STEM Educator |

July. 2025 - Present

- Coordinated and supported the delivery of STEM education programs and technical exhibitions, promoting hands-on learning and innovation.
- Led and contributed to the development of impactful, solution-oriented engineering projects addressing real-world challenges.

BLHCF LAUTECH | Ogbomosho, Oyo State.

Academic Coordinator |

Feb. 2023 - Present

- Coordinated and managed impactful academic tutorials to enhance student learning and engagement.
- Provided academic guidance and career mentorship to undergraduate students, supporting their development and success.

AWARDS AND RECOGNITIONS

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|---|-----------------------------------|
| ● Dean's List Award - Perfect GPA (5.00) | <i>2023/2024 academic session</i> |
| ● Top 1% - Electronic Engineering | <i>2023/2024 Academic Session</i> |
| ● Top 1% - Control Systems and Engineering | <i>2024/2025 Academic Session</i> |
| ● Best Intern - Embedded Systems, Computer Vision and ROS | <i>Elord Technology, 2025</i> |
| ● Scholarship of Excellence - Oluwasedago Foundation | <i>2025</i> |
| ● Co-founder - IEEE LAUTECH Student Branch | <i>2025</i> |
| ● Grant Recipient (\$1,700) - IEEE Region 8 Sustainable Universities Program | <i>2026</i> |
| ● Scholarship Recipient - IEEE Computer Society WEIGD Student Support Fund | <i>2026</i> |

ACADEMIC CONFERENCES/SEMINAR

- Attended African Engineering and Technology Network, Conference titled “**Empowering the Next Generation of African Research Leaders in Digital World**” at University of Lagos, Nigeria, 28th May, 2025

PROFESSIONAL DEVELOPMENT AND CERTIFICATIONS

Advanced Computer Vision and DeepLearning, Udemy *2025*

- Advanced image processing and deep learning model training using PyTorch and TensorFlow, including CNN theory and applications.
- Practical experience with transfer learning, fine-tuning pretrained models, object detection frameworks (YOLOv8, Faster R-CNN) and data preprocessing pipelines.

Mastering Pytorch: 100 project Bootcamp Training, Udemy *2025*

- Completed 100 projects covering CNN model training, transfer learning and LLM applications using PyTorch.

Edge Avoiding Robot Using ROS, Mechamind *2025*

- Learned ROS fundamentals, including writing nodes, publishing/subscribing, URDF creation, robot simulation in Gazebo and sensor-based edge avoidance implementation.

AI Agents Using Langchain and Langraph, Huggingface *2025*

- Learned to integrate AI agents into embedded systems, computer vision pipelines and ROS for intelligent autonomous system development.

VOLUNTEERING AND PROFESSIONAL MEMBERSHIP

Volunteering & Coordination Roles

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|---|------------------------------------|
| ● Project Coordinator - IEEE LAUTECH Student Branch | <i>2025 - Present</i> |
| ● Industrial Activities Board (IAB) - IEEE Robotics and Automation Society (RAS) | <i>2026 - Present</i> |
| ● Executive - Robotics and Artificial Intelligence Nigeria (RAIN) | <i>2025 - Present</i> |
| ● Academic Coordinator - BLHCF LAUTECH | <i>2025 - Present</i> |
| ● Low Budget Engineer | <i>ss</i>
<i>2025 - Present</i> |

Professional Memberships

- Member - Society of Petroleum Engineers (SPE), | ID: 5743743 *2024 - Present*

- Member - International Association of Engineers (IAENG) *2024 - Present*
- Member - Institute of Electrical and Electronic Engineers | ID: 100929513 *2025 - Present*
- Member - Robotics and Artificial Intelligent Nigeria(RAIN) *2025 - Present*